

Experiment specification: scarce-label adaptation with independent evaluation

Draft v0.3 | 2026-09-23 | Two pages | Model experiments remain on hold

Page 1 - question, evidence and independent data pools

Question. At small label budgets, does benign domain-matched adaptation improve detection more than decision-threshold adjustment or generic additional data? Evaluate discrimination, operational false alarms and third-domain retention separately. This is a controlled extension, not a new architecture. Published EvoShield experiments do not report the proposed three-way equal-label-budget comparison (see resolution memo); author confirmation is pending. General domain adaptation is already studied.

Primary collections and domains. Rebuild around PromptShield and InjecGuard, not deepset. PromptShield's pinned test contains 17,030 benign/6,486 positive rows before cleaning; InjecGuard's pinned train contains 61,069/15,666. Their collections are not independent domains: 479 PromptShield test rows exactly match InjecGuard training. Proposed S is generic instruction/conversation tasks; T is natural-language document/application tasks from identified Dolly/Natural-Instructions components; U is code tasks from identified SPP components. Recover row/parent/template IDs and component rights before assigning domains. U is never adapted. These assignments remain provisional; the public PromptShield schema lacks origin fields. Deepset's 627 records are legacy diagnostics only.

License/provenance gate. Both merged pools are on HOLD. Apache/MIT outer declarations do not settle LMSYS transfer restrictions, Alpaca non-commercial terms, Dolly share-alike terms, nested mixtures or missing component permissions. Exclude unresolved sources; recount usable groups afterward. Exclude deepset-derived rows from primary pools. The component audit lists every InjecGuard source. No permission or identity is inferred from a mirror or stylistic similarity.

Separate pools. Freeze source training, source-only calibration, generic addition bank, target addition bank and final S/T/U evaluation groups. Evaluation size is independent of B and uses all permitted independent records available; no 200-row test cap. Use a large separate source calibration pool, with a planning target of at least 5,000 benign items. No extra target calibration labels are available outside B. Evaluation-label review is separately costed and never feeds model selection. Globally group parent documents, variants and templates; add lexical and human-reviewed embedding candidates before partitioning. Inspect licenses and provenance before extending the semantic audit to the primary corpora.

Labels and people. Proposed annotators: student plus supervisor-nominated independent reviewer; senior adjudicator. Reviewer/adjudicator are not assigned yet. Use task/context-aware injection, benign and ambiguous labels, blind to predictions/upstream labels. Do not equate harmfulness, direct jailbreaks or imperatives with indirect injection. Time 20 calibration items; plan 1-2 minutes for short records and 3-5 for long context. Independently double-label at least 100 target audit items (50 upstream-positive/50 benign), plus source/third-domain audits; report pre-adjudication agreement/kappa and 95% group-bootstrap intervals. IAA is currently NOT MEASURED.

Adaptation budget and arms. Full budget catalog: B=0/25/50/100/200 final benign labels revealed to the algorithm. Record actual review/exclusion/adjudication costs separately. Frozen uses no target labels. Threshold-only freezes weights and uses B target labels for its decision threshold. Generic retraining adds B unused S labels. Matched retraining adds the same B target labels as threshold-only. Retraining arms share initialization, updates and sampling weight. All coincide at B=0. Use nested seed-specific subsets from reviewed banks; no hidden free target validation. Equal revealed labels are not automatically equal annotation hours.

Page 2 - power, endpoints, workload and decisions

Model and compact scope. Prespecified primary is source-trained DeBERTa-v3-small, checkpoint pinned before fitting. Freeze hyperparameters/length handling on source-only development. Recommend B=0/50/200 and seeds 17,29,42,71,101; defer 25/100 and extra architectures. One combined linear B=0 diagnostic suffices. Five seeds estimate training/sample sensitivity; overlapping test items are not additional independent observations.

Endpoints. Primary target discrimination: recall at the largest empirical ROC point with FPR $\leq 1\%$, without interpolation. This evaluation-only threshold is never deployed. Threshold-only cannot change ranking, so separately report every arm's recall AND FPR at its allowed calibration-selected operational threshold. Frozen/retrained thresholds use source calibration; threshold-only uses B target benign labels and falls back to source at B=0. Report hard-benign FPR, Brier score, NLL and fixed-bin ECE with intervals and prevalence, plus unchanged-threshold S/U behavior. U is descriptive unless independently powered; no unpowered non-degradation claim.

Explicit power. Confidence and power differ. At true FPR near 1%, approximate negatives for 95% half-widths of 0.5/0.3/0.2 percentage points are 1,522/4,226/9,508. Paired fixed-threshold power depends on discordance q . For a two-point recall difference, $\alpha=.05$ and $q=.10/.20$, exact-McNemar planning gives 2,050/4,025 positives for 80% power; 2,700/5,350 for 90%. With two confirmatory contrasts ($\alpha=.025$), 90% requires 3,175/6,300. These are independent positives, not total rows.

ROC-cutoff uncertainty. Those counts do not power an estimated 1% ROC cutoff. An illustrative Gaussian paired-score calculation with baseline recall .80, correlation .90, 6,500 positives and $\alpha=.025$ requires about 35,212 negatives for 80% power at a two-point effect, versus 3,518 at five points. These assumptions are not measured. Preserve all usable evaluation data; target roughly 10,000 negatives/5,000 positives if available, then assess the detectable effect after grouping. Do not promise a two-point ROC result from that target or from raw corpus counts.

Inference and decisions. At B=200, compare matched with threshold-only and generic using paired group/seed bootstrap (10,000 replicates; seed 20260923). Use 97.5% intervals for these two prespecified contrasts to maintain familywise 95% coverage; report ordinary 95% intervals and all seed results too. Resample complete independent groups, preserving pairs; recompute empirical ROC cutoffs, keep operational thresholds frozen, and report undefined replicates. Conditional McNemar calculations omit between-seed variance; a source-only pilot must check it. Statistical superiority means an adjusted interval above zero. Assess the two-point practical margin separately: an upper bound below two points falsifies a benefit of that size; crossing it is inconclusive. Do not demand a lower bound above five points while claiming power for a true five-point effect.

Run and labor budget. Full grid: $4 \times 5 \times 10 \times 3 = 600$ nominal evaluation cells, but shared baselines/threshold score reuse reduce it to 90 fits and 130 unique configurations. Compact: 180 nominal cells, 25 fits, 35 configurations, 75 model-domain scoring passes. Hypothetical compact training needs 15-50 GPU-hours before scoring/contingency; runtime and hardware are UNVERIFIED. Annotating ≤ 800 adaptation candidates plus 300 evaluation-audit records with two reviewers at two minutes/judgment costs about 86.7 person-hours including assumed adjudication/admin; four minutes raises it to 160. Public evaluation labels are reused with a documented audit, not all relabeled.

Eight-week gates. W1: literature, permissions and power. W2: provenance recovery, named reviewers, timed pilot and target IAA ≥ 100 . W3: banks, full-corpus semantic/provenance grouping and independent counts. W4: only after gates, source-only runtime/variance pilot and protocol freeze. W5: compact fits. W6: one locked evaluation. W7: analysis/report. W8: reproduction/revision. If staffing/provenance is unresolved after W2 or counts fail the chosen power design after W3, narrow to exploratory replication/data audit. Do not cut evaluation size to retain more runs.

Supporting records: EVOSHIELD_RESOLUTION.md; COMPONENT_PROVENANCE_AUDIT.md; POWER_ANALYSIS.md; ANNOTATION_PLAN.md; SEMANTIC_SENSITIVITY.md; EIGHT_WEEK_SCOPE.md. No detector training or adaptation was run for this revision.